DAY 35 LAB: Extended ACLs

Goals:

1. Configure extended ACLs to fulfill the following network policies:
    a. Hosts in 172.16.2.0/24 can't communicate with PC1. **[DONE]**
    b. Hosts in 172.16.1.0/24 can't access the DNS service on SRV1. **[DONE]**
    c. Hosts in 172.16.2.0/24 can't access the HTTP or HTTPS services on SRV2. **[DONE]**

Important commands that are used:

Full form command	Purpose
router ospf 1	To enter router configuration for OSPF and its process ID
passive-interface g0/0	Used for end hosts intermediary devices to disable receiving “OSPF Hello messages (useful to find potential router neighbors)” unnecessary traffic for end hosts.
network 192.168.1.0 0.0.0.255 area 0	To enable OSPF in this network and specify its area
ip access-list extended 1	To create the numbered standard access-list
Ip access-list extended FOR-192.168.1.0/24	To create the named standard access-list
10 {deny | permit} tcp 192.168.1.0 0.0.0.255 host 1.1.1.1 eq 80	To create specific ACE (Access Control Entry) in subconfig mode. In this example you specified the sequence number, action, protocol keyword, source network, wildcard mask, destination host, and destination port. This means that you either deny/permit the network 192.168.1.0/24 to access the HTTP service in host/server 1.1.1.1
Access-list {sequence-number} {deny | permit} tcp 192.168.1.0 0.0.0.255 host 1.1.1.1 eq 443	To create ACE (Access Control Entry) in **numbered extended** ACL. In this example you create the ACL and ACE in global config mode.

remark “PUT YOUR REMARKS HERE”	Description how this ACL work
ip access-group 1 { out | in }	Applying the ACL in the interface router (numbered extended ACL way).
ip access-group FOR-192.168.1.0/24 { out | in }	Applying the ACL in the interface router (named extended ACL way)
Lagay mo dito kung paano mo ginawa yung sa DNS at HTTP skibidis	
nslookup www.njl.com	To know the IP address of this “www.njl.com” and test whether the DNS name resolution is working (coverts a domain name into an IP address).
Ping 192.168.1.100 (DNS ip address)	To test the connectivity between DNS and client

What I have learned:

→ How to create standard { numbered | named } access control list (ACLs).
→ How to create access control entries (ACE) for access control list (ACLs).
→ How to apply the access control list (ACLs) to interface
→ How to setup set-up the DNS server and create a DNS on it
→ How to verify the DNS I create (In browser and ping) and modify the HTML in HTTPS/HTTP section of the server option.

Verification:

→ Hosts in 172.16.2.0/24 can't communicate with PC1
→ Hosts in 172.16.1.0/24 can't access the DNS service on SRV1.
→ Hosts in 172.16.2.0/24 can't access the HTTP or HTTPS services on SRV2
→ The DNS server for 192.168.1.0/24 and 192.168.2.0/24 are working by testing other network that are not affected by applied ACLs